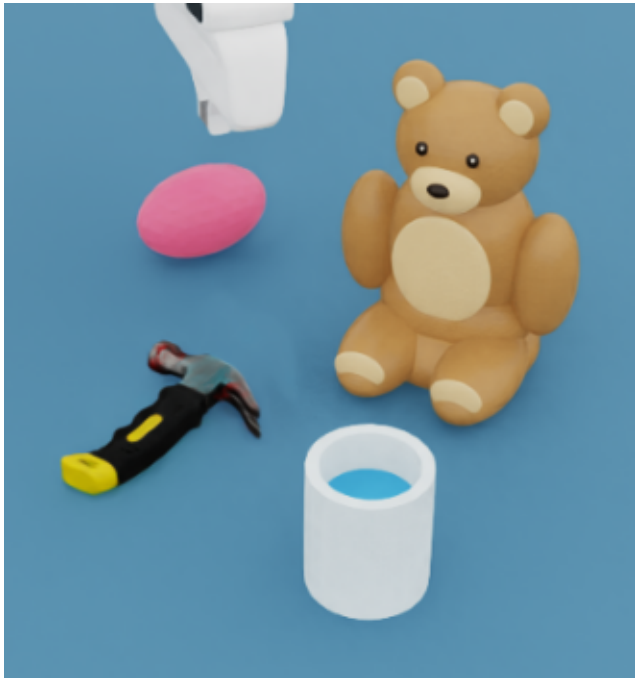
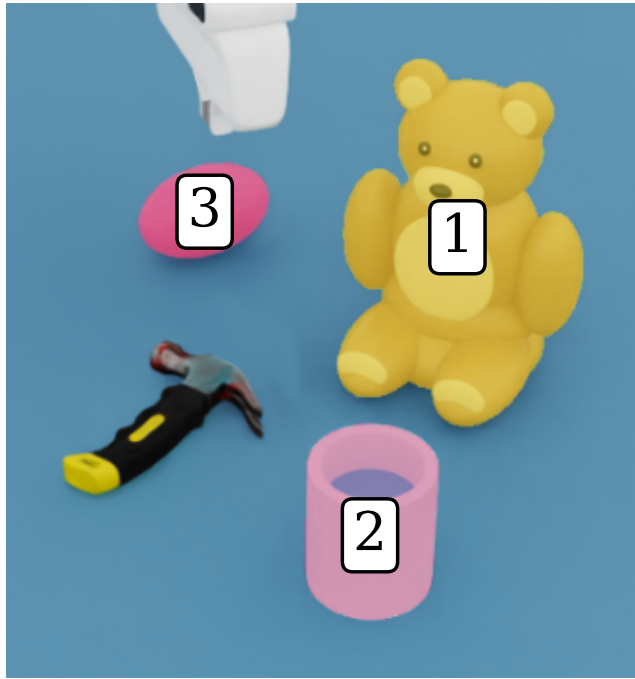


Original scene I



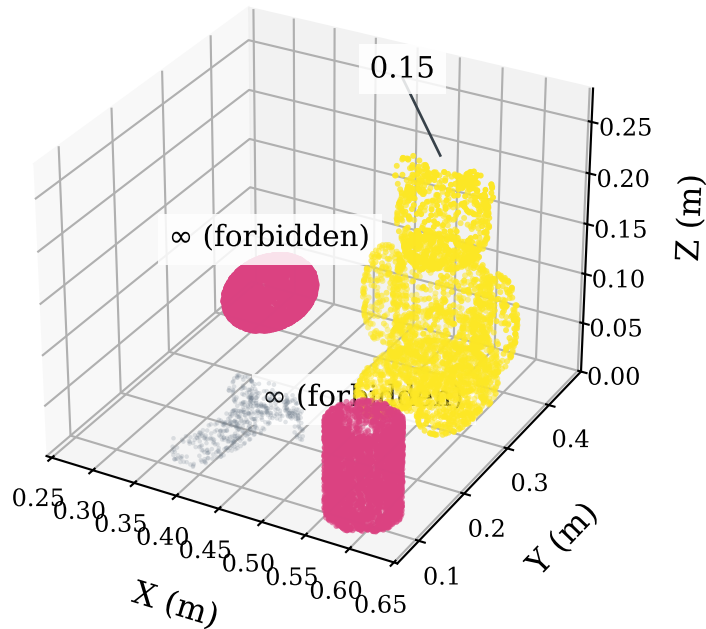
Annotated scene I'



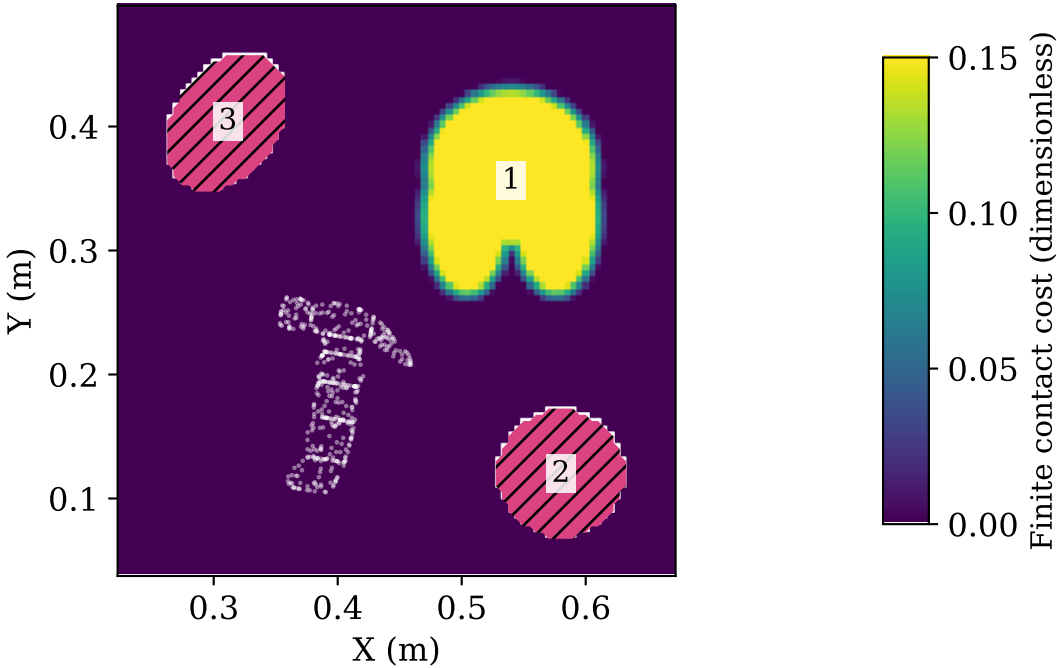
Assistant review
for this prototype

**Vision
Language
Model**

3-D contact cost $C(p)$



XY slice at $Z = 0.015$ m



Forbidden: $C = \infty$ (separate from finite color scale)

Safely extract the hammer.
(1) Doll: contact is acceptable.
(2) Liquid cup and (3) balloon: avoid contact.

Language instruction ℓ

3-D point query: 5 mm grid; 12 mm clearance band.
Geometry: recorded simulation meshes and poses.